LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all

a) Create a MapReduce program to find average temperature for each year from NCDC data set.
b) find the mean max temperature for every month.

package temp;

CODE, COMMAND WITH OUTPUT – A

Driver Code

```
import
org.apache.hadoop.conf.Config
uration; import
org.apache.hadoop.fs.Path;

import
org.apache.hadoop.io.IntWrit
able; import
org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import
org.apache.hadoop.mapreduce.lib.input.FileIn
putFormat; import
```

```
org.apache.hadoop.mapreduce.lib.output.File
OutputFormat;

public class AverageDriver {
    public static void main(String[] args)

      throws Exception { if (args.length !=

      2) {

      System.err.println("Please enter both input and
      output parameters."); System.exit(-1);
    }

    // Creating a configuration and
    job instance Configuration conf =
    new Configuration();
    Job job = Job.getInstance(conf, "Average

    Calculation");

    job.setJarByClass(AverageDriver.class);

    // Input and output paths
    FileInputFormat.addInputPath(job, new
    Path(args[0]));
    FileOutputFormat.setOutputPath(job,
    new Path(args[1]));

    // Setting mapper and reducer
    classes
    job.setMapperClass(AverageMa
    pper.class);
    job.setReducerClass(AverageRe
    ducer.class);

    // Output key and value types
    job.setOutputKeyClass(Text.clas
    s);
    job.setOutputValueClass(IntWrita
    ble.class);

    // Submitting the job and waiting for it to
    complete
    System.exit(job.waitForCompletion(true) ?
    0 : 1);
  }
```

```
}
```

Mapper Code

```
package temp;

import java.io.IOException;

import
org.apache.hadoop.io.IntWrit
able; import
org.apache.hadoop.io.LongW
ritable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

  public class AverageMapper extends Mapper<LongWritable,

    Text, Text, IntWritable> { public static final int MISSING =

    9999;

  @Override
      public void map(LongWritable key, Text
          value, Context context) throws
          IOException, InterruptedException {

    String line = value.toString();

    // Extract year from fixed
    position String year =
    line.substring(15, 19); int
    temperature;

    // Determine if
    there's a '+' sign if
    (line.charAt(87) ==
    '+') {
      temperature = Integer.parseInt(line.substring(88, 92));
    } else {
```

```
        temperature = Integer.parseInt(line.substring(87, 92));
      }

      // Quality check character
      String quality = line.substring(92, 93);

      // Only emit if data is valid
        if (temperature != MISSING &&
          quality.matches("[01459]")) {
          context.write(new Text(year), new
          IntWritable(temperature));
      }
    }
  }
```

Reducer Code

```
package temp;

import java.io.IOException;

import
org.apache.hadoop.io.IntWr
itable; import
org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
  public class AverageReducer extends Reducer<Text,

    IntWritable, Text, IntWritable> { @Override

  public void reduce(Text key, Iterable<IntWritable> values,
              Context context) throws IOException, InterruptedException {

    int
    sumTemp
    = 0; int
    count = 0;

      for (IntWritable value
        : values) {
        sumTemp +=
        value.get();
        count++;
    }

    if (count > 0) {
      int average = sumTemp /
      count; context.write(key, new
      IntWritable(average));
```

```
    }
  }
}
```

```
thilak@kali: ~/weather
Session  Actions  Edit  View  Help
┌──(thilak㉿kali)-[~]
└─$ mkdir weather
cd weather

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageDriver.java

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageMapper.java

┌──(thilak㉿kali)-[~/weather]
└─$ nano AverageReducer.java

┌──(thilak㉿kali)-[~/weather]
└─$ javac -classpath $(hadoop classpath) -d . *.java
hadoop: command not found
error: invalid flag: .
Usage: javac <options> <source files>
use --help for a list of possible options

┌──(thilak㉿kali)-[~/weather]
└─$ export HADOOP_HOME=/home/thilak/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

┌──(thilak㉿kali)-[~/weather]
└─$ javac -classpath $(hadoop classpath) -d . *.java

┌──(thilak㉿kali)-[~/weather]
└─$ jar -cvf weather.jar *
added manifest
adding: AverageDriver.class(in = 1369) (out= 748)(deflated 45%)
adding: AverageDriver.java(in = 919) (out= 374)(deflated 59%)
adding: AverageMapper.class(in = 1708) (out= 697)(deflated 59%)
adding: AverageMapper.java(in = 657) (out= 315)(deflated 52%)
adding: AverageReducer.class(in = 1609) (out= 677)(deflated 57%)
adding: AverageReducer.java(in = 527) (out= 267)(deflated 49%)

┌──(thilak㉿kali)-[~/weather]
└─$ nano weather.txt

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -mkdir /weatherinput
hdfs dfs -put weather.txt /weatherinput/

┌──(thilak㉿kali)-[~/weather]
└─$ hadoop jar weather.jar AverageDriver /weatherinput /weatheroutput
2026-05-02 14:55:24,936 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-05-02 14:55:25,104 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-05-02 14:55:25,104 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-05-02 14:55:25,313 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-02 14:55:25,575 INFO input.FileInputFormat: Total input files to process : 1
2026-05-02 14:55:25,609 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-02 14:55:25,836 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1283312055_0001
2026-05-02 14:55:25,836 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-02 14:55:26,059 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-05-02 14:55:26,062 INFO mapreduce.Job: Running job: job_local1283312055_0001
2026-05-02 14:55:26,071 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-05-02 14:55:26,109 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-02 14:55:26,112 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:55:26,112 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:55:26,117 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-05-02 14:55:26,222 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-05-02 14:55:26,227 INFO mapred.LocalJobRunner: Starting task: attempt_local1283312055_0001_m_000000_0
2026-05-02 14:55:26,280 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-05-02 14:55:26,282 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:55:26,283 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:55:26,341 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-05-02 14:55:26,351 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/weatherinput/weather.txt:0+282
```

```
thilak@kali: ~/weather
Session  Actions  Edit  View  Help
                FILE: Number of write operations=0
                HDFS: Number of bytes read=564
                HDFS: Number of bytes written=22
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=3
                Map output records=3
                Map output bytes=27
                Map output materialized bytes=39
                Input split bytes=111
                Combine input records=0
                Combine output records=0
                Reduce input groups=2
                Reduce shuffle bytes=39
                Reduce input records=3
                Reduce output records=2
                Spilled Records=6
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=15
                Total committed heap usage (bytes)=419430400
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=282
        File Output Format Counters
                Bytes Written=22

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -cat /weatheroutput/part-00000
cat: `/weatheroutput/part-00000': No such file or directory

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -ls /weatheroutput
Found 2 items
-rw-r--r--   3 thilak supergroup          0 2026-05-02 14:57 /weatheroutput/_SUCCESS
-rw-r--r--   3 thilak supergroup         22 2026-05-02 14:57 /weatheroutput/part-r-00000

┌──(thilak㉿kali)-[~/weather]
└─$ hdfs dfs -cat /weatheroutput/part-r-00000
5202    89012
5678    89012

┌──(thilak㉿kali)-[~/weather]
└─$
```